

Denoising Score Matching for the Finite-Dimensional Particle Approximation

1 Setup

Fix $n \in \mathbb{N}$ and a mass vector

$$s = (s_1, \dots, s_n) \in \Sigma_n^{1/n}.$$

We consider particle positions

$$x = (x_1, \dots, x_n) \in (\mathbb{T}^d)^n.$$

Conditionally on the mass vector s , the free n -particle system evolves only in the position variables and satisfies

$$d\xi_i(t) = \sqrt{\frac{2}{s_i}} d\beta_i(t), \quad i = 1, \dots, n,$$

where $\beta_1, \dots, \beta_n$ are independent Brownian motions on $\mathbb{R}^d$, projected modulo $\mathbb{Z}^d$ to $\mathbb{T}^d$. Its generator is

$$L^{(n),s} = \sum_{i=1}^n s_i^{-1} \Delta_{x_i}.$$

We also define the weighted gradient by

$$D_i^{(n)} = \frac{1}{s_i} \nabla_{x_i}, \quad D^{(n)}u = (D_1^{(n)}u, \dots, D_n^{(n)}u).$$

In the theoretical h -transform construction, the drifted dynamics involve

$$2D^{(n)} \log u_t^{(n)}, \quad u_t^{(n)} = P_{T-t}^{(n)} g_n,$$

where $P_t^{(n)}$ is the free semigroup and g_n is a terminal weight.

2 Three-step denoising score matching procedure

We now describe a denoising score matching interpretation of this finite-dimensional model.

Step 1: data representation and forward noising

Let $\rho_{0,s}$ denote the law of the clean particle configuration

$$X = (X_1, \dots, X_n) \in (\mathbb{T}^d)^n$$

conditional on the mass vector s .

For a noise level $\tau > 0$, define the noisy configuration

$$Y = (Y_1, \dots, Y_n)$$

by running the free particle dynamics for time τ starting from X . Equivalently, the conditional law of Y given $X = x$ is the transition kernel

$$p_{\tau,s}(y \mid x).$$

The corresponding noisy marginal density is

$$\rho_{\tau,s}(y) := \int_{(\mathbb{T}^d)^n} p_{\tau,s}(y \mid x) \rho_{0,s}(x) dx.$$

In semigroup notation,

$$\rho_{\tau,s} = P_{\tau}^{(n),s} \rho_{0,s}.$$

Step 2: learn the score field

The object needed for reverse-time generation is the score

$$D_y^{(n)} \log \rho_{\tau,s}(y).$$

A score network is a map

$$S_{\theta}(\tau, s, y) \in (\mathbb{R}^d)^n$$

trained to approximate this quantity.

A natural weighted least-squares loss is

$$\mathcal{L}(\theta) = \mathbb{E} \left[\sum_{i=1}^n s_i \|S_{\theta,i}(\tau, s, Y) - T_i(\tau, s, X, Y)\|^2 \right],$$

where the target is chosen to be

$$T(\tau, s, X, Y) = D_y^{(n)} \log p_{\tau,s}(Y \mid X).$$

Step 3: reverse-time generation

After training, one samples an initial noisy state from the reference law

$$s \sim \pi_n^{\beta}, \quad Y_T \sim m^{\otimes n},$$

and then runs a reverse-time sampling scheme in which the unknown exact drift

$$2D^{(n)} \log u_t^{(n)}(s, y)$$

is replaced by the learned approximation

$$2S_{\theta}(T - t, s, y).$$

Thus the denoising score matching construction attempts to learn directly the same quantity that appears in the finite-dimensional h -transform drift.

3 Why the theoretical target is a conditional expectation

We now show that the score of the noisy marginal can be written as a conditional expectation of the score of the forward kernel.

Fix $s \in \Sigma_n^{1/n}$ and define

$$\rho_{\tau,s}(y) = \int p_{\tau,s}(y | x) \rho_{0,s}(x) dx.$$

Assume enough regularity to differentiate under the integral sign. Then

$$D_y^{(n)} \rho_{\tau,s}(y) = \int D_y^{(n)} p_{\tau,s}(y | x) \rho_{0,s}(x) dx.$$

Using

$$D_y^{(n)} p_{\tau,s}(y | x) = p_{\tau,s}(y | x) D_y^{(n)} \log p_{\tau,s}(y | x),$$

we obtain

$$D_y^{(n)} \rho_{\tau,s}(y) = \int p_{\tau,s}(y | x) D_y^{(n)} \log p_{\tau,s}(y | x) \rho_{0,s}(x) dx.$$

Dividing by $\rho_{\tau,s}(y)$ gives

$$D_y^{(n)} \log \rho_{\tau,s}(y) = \frac{\int p_{\tau,s}(y | x) D_y^{(n)} \log p_{\tau,s}(y | x) \rho_{0,s}(x) dx}{\int p_{\tau,s}(y | x) \rho_{0,s}(x) dx}.$$

By Bayes' rule,

$$\mathbb{P}(X \in dx | Y = y, s) = \frac{p_{\tau,s}(y | x) \rho_{0,s}(x)}{\rho_{\tau,s}(y)} dx.$$

Hence

$$D_y^{(n)} \log \rho_{\tau,s}(y) = \mathbb{E} \left[D_y^{(n)} \log p_{\tau,s}(y | X) \mid Y = y, s \right].$$

Therefore, if one minimizes the loss

$$\mathcal{L}(\theta) = \mathbb{E} \left[\sum_{i=1}^n s_i \| S_{\theta,i}(\tau, s, Y) - D_{y_i}^{(n)} \log p_{\tau,s}(Y | X) \|^2 \right],$$

then the population minimizer is exactly

$$S_{\theta}^*(\tau, s, y) = D_y^{(n)} \log \rho_{\tau,s}(y).$$

4 Calculation of the score of the forward kernel

We now compute the score of the forward kernel associated with the free particle system.

4.1 Euclidean cover

On the universal cover $(\mathbb{R}^d)^n$, the conditional law of Y given $X = x$ is a product Gaussian:

$$p_{\tau,s}^{\mathbb{R}^{dn}}(y | x) = \prod_{i=1}^n \left(\frac{s_i}{4\pi\tau} \right)^{d/2} \exp \left(-\frac{s_i}{4\tau} |y_i - x_i|^2 \right).$$

Therefore

$$\log p_{\tau,s}^{\mathbb{R}^{dn}}(y | x) = C(\tau, s) - \sum_{i=1}^n \frac{s_i}{4\tau} |y_i - x_i|^2,$$

where $C(\tau, s)$ is independent of y .

Differentiating with respect to y_i yields

$$\nabla_{y_i} \log p_{\tau,s}^{\mathbb{R}^{dn}}(y | x) = -\frac{s_i}{2\tau}(y_i - x_i).$$

Applying the weighted gradient operator $D_i^{(n)} = \frac{1}{s_i} \nabla_{y_i}$, we get

$$D_{y_i}^{(n)} \log p_{\tau,s}^{\mathbb{R}^{dn}}(y | x) = -\frac{1}{2\tau}(y_i - x_i).$$

Hence the full weighted kernel score is

$$D_y^{(n)} \log p_{\tau,s}^{\mathbb{R}^{dn}}(y | x) = \left(-\frac{y_1 - x_1}{2\tau}, \dots, -\frac{y_n - x_n}{2\tau} \right).$$

4.2 Torus case

On the torus $(\mathbb{T}^d)^n$, the exact transition kernel is the product heat kernel

$$p_{\tau,s}(y | x) = \prod_{i=1}^n p_{\tau/s_i}^{\mathbb{T}^d}(y_i, x_i),$$

where $p_t^{\mathbb{T}^d}$ denotes the heat kernel on $\mathbb{T}^d$ corresponding to the equation

$$\partial_t u = \Delta u.$$

Thus

$$\log p_{\tau,s}(y | x) = \sum_{i=1}^n \log p_{\tau/s_i}^{\mathbb{T}^d}(y_i, x_i),$$

and therefore

$$\nabla_{y_i} \log p_{\tau,s}(y | x) = \nabla_{y_i} \log p_{\tau/s_i}^{\mathbb{T}^d}(y_i, x_i).$$

Applying the weighted gradient again gives

$$D_{y_i}^{(n)} \log p_{\tau,s}(y | x) = \frac{1}{s_i} \nabla_{y_i} \log p_{\tau/s_i}^{\mathbb{T}^d}(y_i, x_i).$$

For small τ , or whenever wrap-around effects are negligible, this is well approximated by the Euclidean expression

$$D_{y_i}^{(n)} \log p_{\tau,s}(y | x) \approx -\frac{1}{2\tau}(y_i - x_i).$$

5 Relation to the theoretical drift

The theoretical finite-dimensional drift has the form

$$2D^{(n)} \log u_t^{(n)}, \quad u_t^{(n)} = P_{T-t}^{(n)} g_n.$$

If we introduce the forward time variable

$$\tau = T - t$$

and set

$$\rho_{\tau,s} = u_t^{(n)}(s, \cdot),$$

then the relevant score is

$$D^{(n)} \log \rho_{\tau,s} = D^{(n)} \log u_t^{(n)}.$$

The denoising score matching construction is therefore aligned with the theory in the following sense:

- (1) the forward noising process is exactly the free particle semigroup;
- (2) the theoretical target is the score $D^{(n)} \log u_t^{(n)}$ of the heat-smoothed density;
- (3) this score can be learned by regressing against the explicit conditional kernel score

$$D_y^{(n)} \log p_{\tau,s}(Y \mid X),$$

because

$$D_y^{(n)} \log \rho_{\tau,s}(y) = \mathbb{E} \left[D_y^{(n)} \log p_{\tau,s}(y \mid X) \mid Y = y, s \right].$$